

Converge Hackathon

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By
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Project objective:

Analyze 211 calls and referral processes to identify gaps and points of referral drop-off

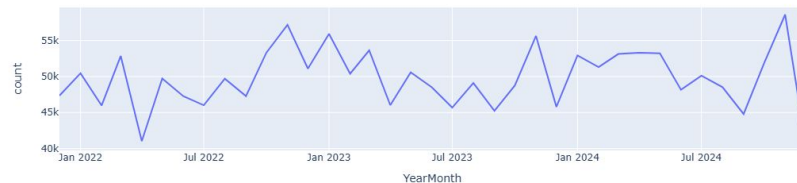
Background Research/ Motivation

- Emerging adults, particularly those from marginalized groups, face complex challenges and often rely on informal supports to access mental health services, but poor coordination across health, education, and community sectors significantly hinders continuity of care and service navigation.
- The absence of real-time, centralized information about service availability and eligibility contributes to long wait times and prevents effective navigation across services. Waitlists, averaging 303 days and in some cases up to five years, exacerbate these challenges
- Referrals are often "cold hand-offs," where individuals are provided with contact information but must independently pursue services without any formal linkage or transfer of information between organizations. This increases the risk of drop-offs, especially among vulnerable populations
- In the Emerging Adult Final Report, a key issue is the poor coordination between healthcare, community, and education services. Each operates largely in isolation, creating barriers to seamless care and complicating navigation for clients
- The mental health system is structured in a way that fosters gaps in timely, consistent, and coordinated care. Providers often rely on informal relationships rather than standardized referral systems, which results in incomplete information and inconsistent experiences for clients

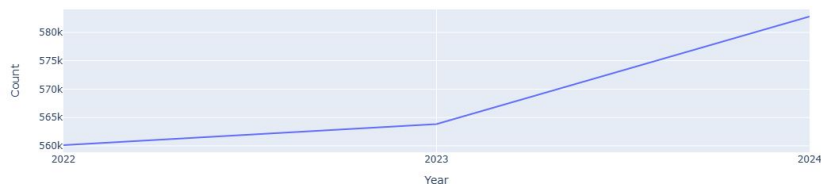
Overview 211 Canada

- 1,014,962 unique calls received over 4 years
- There is a modest increase in calls received each year since 2021.
- Most needs came in through calls (91.5%), followed by chat (4.43%).
- Top 3 provinces with needs are Ontario, Alberta and British Columbia.
- Calls are received primarily to request information, referral or assessment/eligibility for services.
- An unmet need is defined as a call where the caller declines the referral or information provided.
- Out of all calls received in total only 4.7% of calls led to unmet needs.

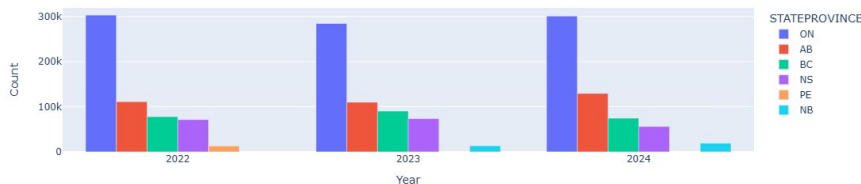
211 Call Volume Canada

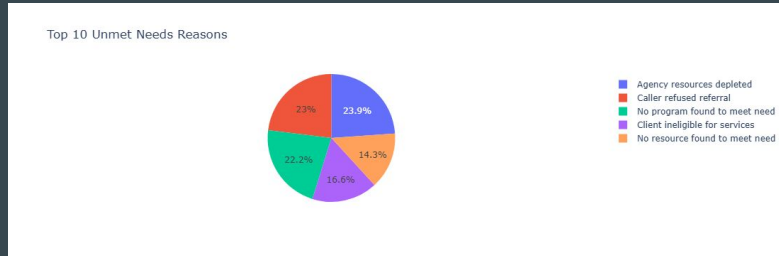
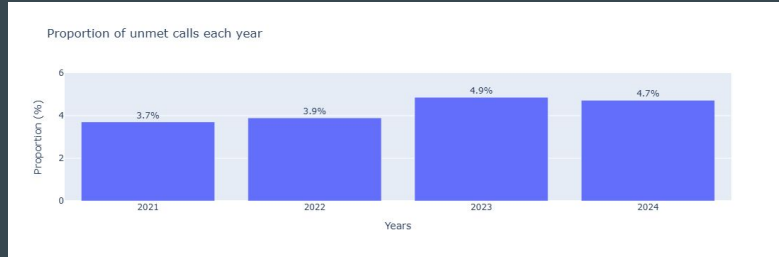


Yearly Call Volumes (>=2 mins length)



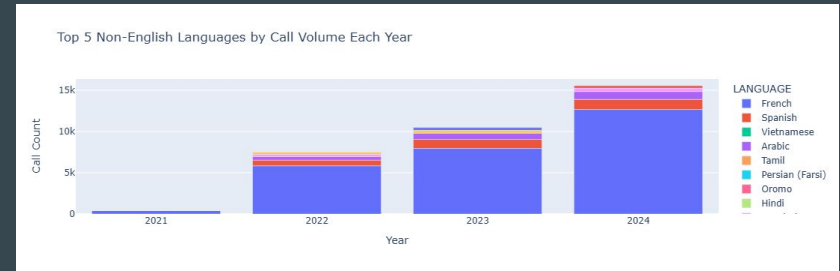
Provincial Call Volume





- Looking at yearly trends, we see there is an increase in proportion of unmet needs.
- Common reasons for unmet needs include
 - Agency resource depleted (23.9%)
 - Caller refused referral (23%)
 - No Program found (22.2%)
 - Client in eligible for services (16.6%)
- Furthermore, looking at calls with ≥ 2 needs:
 - The total number of callers in 2022 with 2 or more needs: 121104 (20.46%)
 - The total number of callers in 2023 with 2 or more needs: 117406 (19.72%)
 - The total number of callers in 2024 with 2 or more needs: 123111 (20.29%)

- Another key factor to consider is the increase in immigration and subsequent increase in demand for social services.. Given Canada's diverse population, language may act as a proxy for identifying immigrants and newcomers, though it is not a definitive indicator of legal status.
- Although Quebec had the lowest number of calls in our dataset, there has been a noticeable rise in calls from non-English speakers, with French being the most common among them.
- While language alone cannot reliably indicate immigration status, the increase in non-English calls — particularly outside of Quebec — suggests the need to explore whether certain unmet needs are disproportionately affecting newcomers and immigrant communities
- Their common needs are related to housing, individual/family services, income support services and citizenship/immigration services.



Understanding the problem

Increased unmet needs

With rise in immigration and the need for social services, the number of unmet needs are likely to increase.

Referral Pipeline

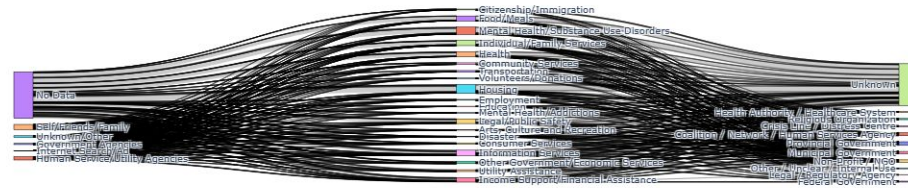
The absence of an established referral pipeline places the burden on callers to follow up and repeatedly share their stories — a process that can be retraumatizing and disempowering.

Lack of Data and connection

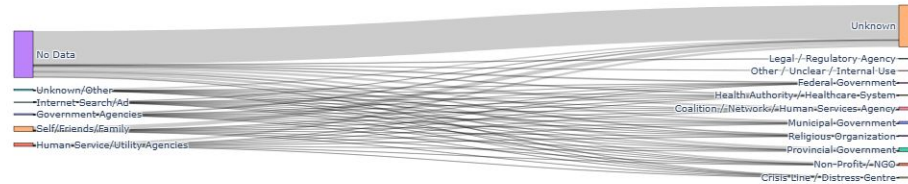
Incomplete or inconsistent data hinders accurate assessment of needs and service gaps, while poor coordination between agencies leads to resource inefficiencies and missed opportunities for effective care.

Sankey Diagram to map flow

Referral Flow Sankey Diagram



Referral Flow: Intake Group to Output Group



Evaluation Summary

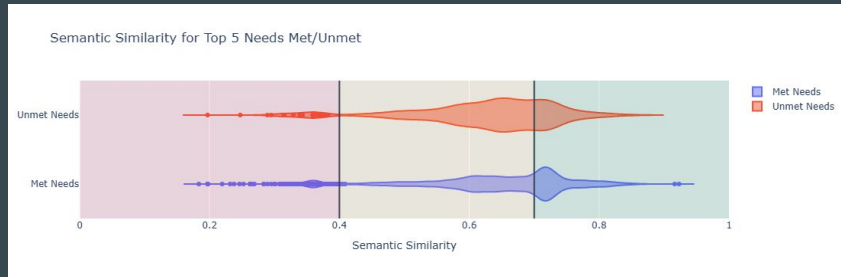
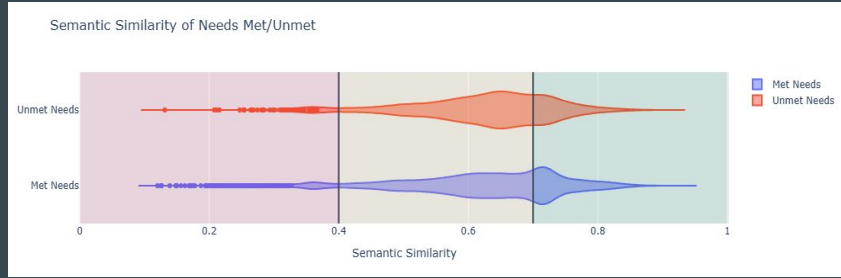
The current referral process relies on soft handoffs, where 211 provides information but no formal transfer of client details occurs between agencies. This lack of coordinated intake forces callers to repeat their stories, potentially retraumatizing them and discouraging follow-through.

Without a structured referral pipeline, vulnerable individuals risk falling through the cracks and missing the care they need and deserve.

Assessing Referrals using NLP: SBERT

SBERT Overview:

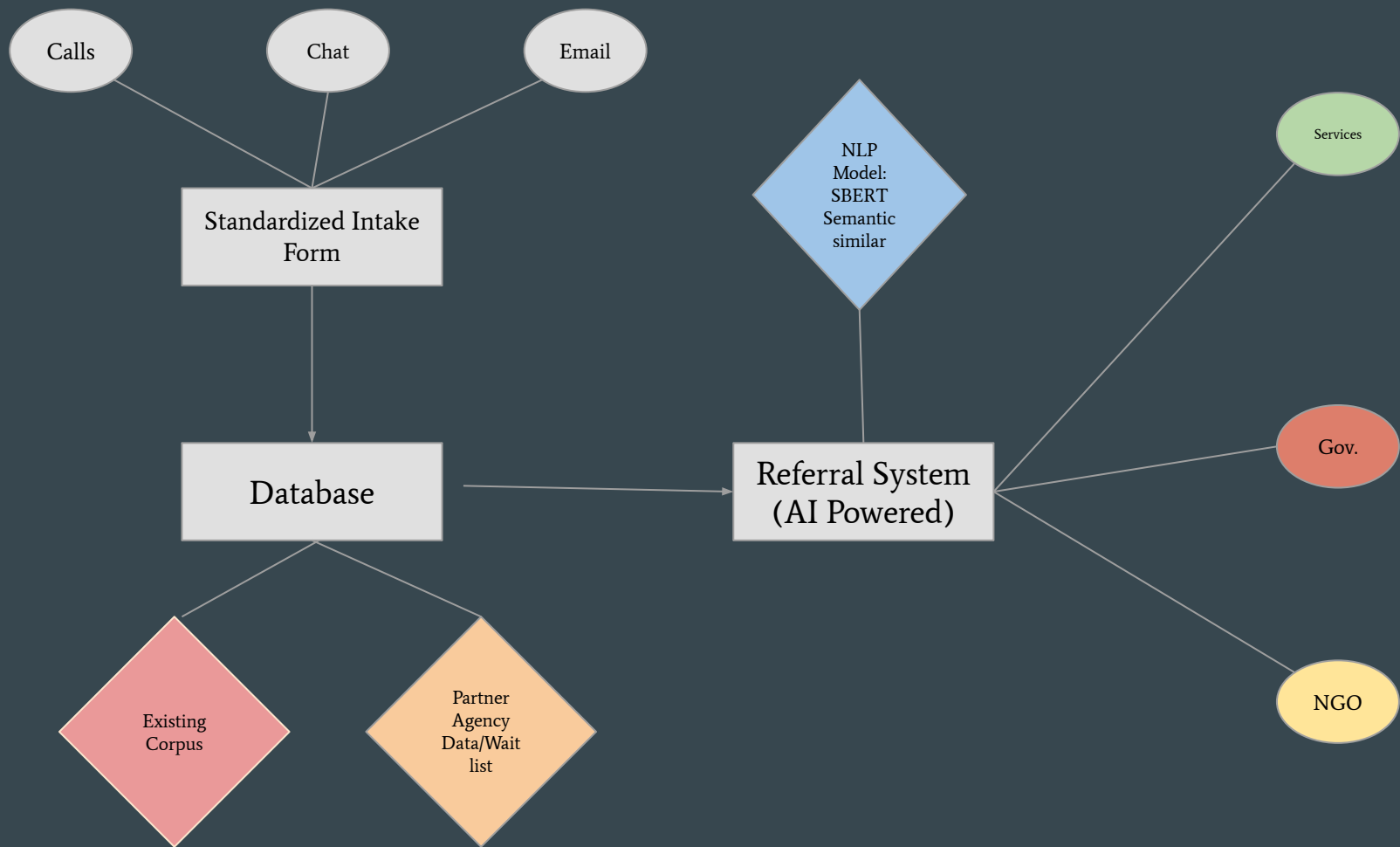
- Utilizing Natural Language Processing machine learning models we can assess the quality of the referrals made.
- By combining columns in the Canada 211 datasets, into a single text value, we can determine the quality of referral made.
- We aggregate key fields (AIRS need category, Levels 1–5, call type) into a single text string per call and likewise combine the referred-Service description.
- Using Sentence-Transformers, we encode these texts into semantic vectors and compute cosine similarity (–1 to 1) as our quality metric.
- Cosine similarity values range in values from -1 to 1, with 1 indicating a strong positive match and -1 indicating the opposite. We set the benchmark for a good referral to be any value ≥ 0.70 .
- The primary focus was to study two sub-groups within our dataset, Met/Unmet needs, though this technique can be utilized to study any sub groups of interest
- The benchmark value for a ‘successful’ referral was ≥ 0.7 , with values in between 0.4-0.7 as good needing improvement and < 0.4 to be an ‘unsuccessful’ referral.



- The study involved collecting samples from each sub group, Met needs had 40,000 samples and Unmet needs had 5,900 samples
- Upon visual inspection we can see that there isn't any major differences between the two groups, however the need met group does show a slight bias towards higher values.
- Unmet needs exhibited marginally lower semantic similarity scores, while met needs showed a modest high-end uplift. However, neither the Z-test ($z = 0.0823$) nor Welch's t-test ($t = 0.0882$) detected a significant difference between the groups.
- Focusing on the top five needs reveals a statistically significant difference in referral quality (Z-test $p = 2.22 \times 10^{-16}$; Welch's t-test $p = 2.50 \times 10^{-16}$). Although the shift is modest, the violin plot's right skew confirms meaningful variation.

Proposed Solution

- **Unified AI-Driven Intake & Referral Platform** ; A single, standardized intake form captures consistent, reliable data in one streamlined step, powering an AI-enabled referral system.
- **NLP-Powered Needs Assessment** ; Models trained on HSIS taxonomy and existing corpora automatically classify and match caller needs to agency waitlists, improving accuracy as more partners join.
- **Privacy-Compliant Data Standardization** ; Anonymous, governed data collection creates a valid operational dataset, ensuring compliance with data privacy and governance policies.
- **Real-Time Tracking & Feedback;** Callers and agencies can monitor referrals and waitlist status, while feedback loops drive continuous system enhancements.
- **Scalable, Region-Aware Optimization;** The pipeline can expand downstream for health-quality evaluation and apply region- and season-specific adjustments for maximum flexibility and impact.



References:

- Canadian Mental Health Association (CMHA). Emerging Adult Mental Health: Final Report. CMHA, 2025. PDF.
- **Converge Mental Health Coalition.** *Converge: Final Report.* Converge Mental Health Coalition, 2025. PDF.